AXON ERP · ARCHITECTURE SERIES

Enterprise Workflow & Cross-Module Integration Specification

Complete operational manual, visual vector flowcharts, state machine engines, and cross-module data connection blueprints across all 18 enterprise domains.

PLATFORM VERSION
v2.4.0 Enterprise Edition

DOCUMENT CLASSIFICATION
Confidential · System Architecture Reference

CORE ARCHITECTURE
FastAPI · PostgreSQL · React · Redis

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Executive Summary & Table of Contents

Axon ERP is an enterprise-grade manufacturing and supply chain operating system. It coordinates the end-to-end commercial lifecycle across multi-site manufacturing facilities, automated warehouses, regulatory compliance gateways, and double-entry financial ledgers.

This technical manual provides an exhaustive operational reference for every domain module in the platform. Each section pairs a **high-fidelity vector flowchart** with precise step-by-step business logic, state machine specifications, and explicit upstream and downstream cross-module integration contracts.

DOCUMENT STRUCTURE & NAVIGATION

SEC 0 **Global Enterprise Architecture & Value Stream Flowchart** — Global Supply Chain Topology

MOD 01 **Identity, Authentication & Role-Based Access Control (RBAC)** — /login, /users, /profile

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MOD 13 **Warehouse Wave Picking, Packing & Cartonization** — /wave-picking, /wms-mobile

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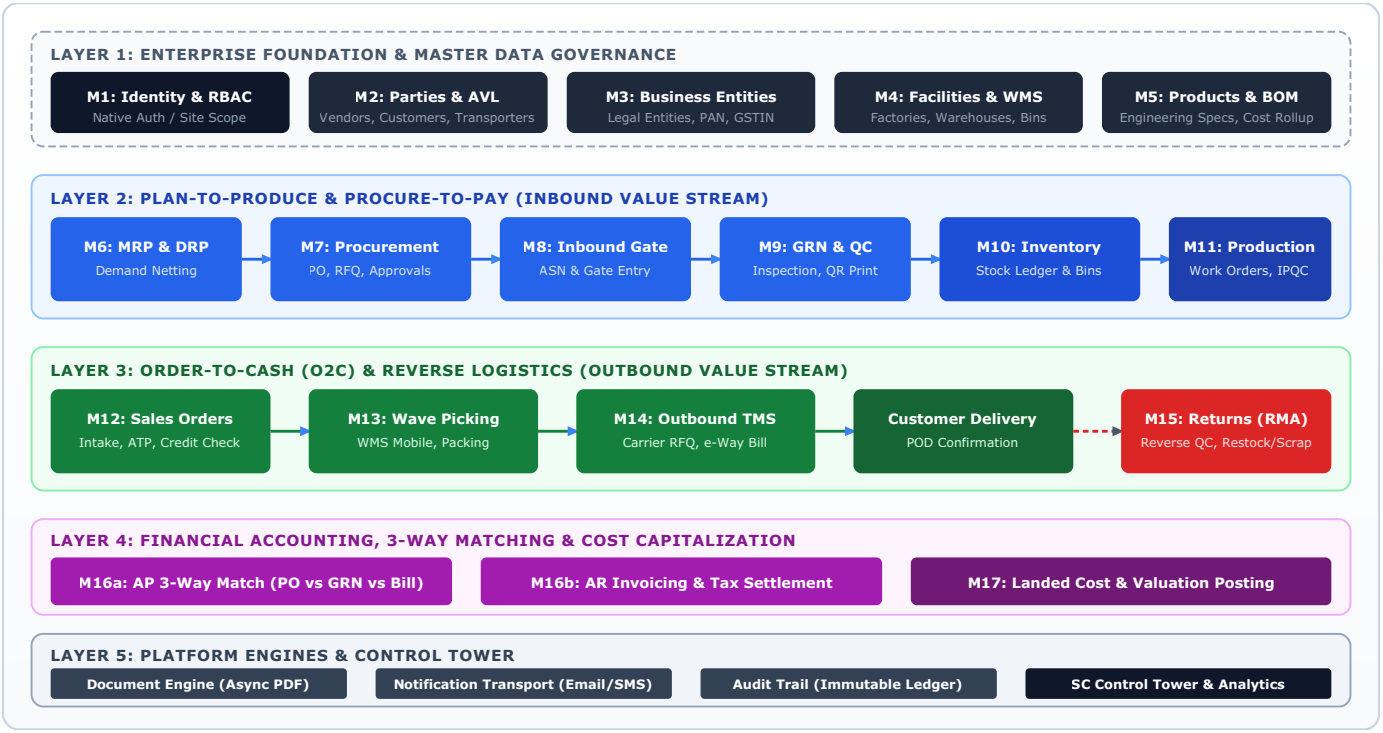
SEC 19 **Master Cross-Module Dependency & Integration Matrix** — End-to-End Value Streams

ARCHITECTURAL PRINCIPLES · THE MODULAR MONOLITH ENGINE

Axon ERP is engineered as a clean *modular monolith* with strictly bounded domains. State transitions in one module (such as GRN approval) trigger transactional side effects in dependent modules (such as Inventory balance increases and Finance 3-way match staging) through domain service contracts and ACID database transactions, preventing data desynchronization.

Global Enterprise Architecture & Value Stream

The diagram below depicts the end-to-end operational lifecycle of Axon ERP. Data originates in the foundational governance layers and cascades through procurement, manufacturing, warehousing, order fulfillment, and financial accounting.



INBOUND STREAM · PROCURE-TO-PAY (P2P)

1. **Demand:** Sales orders & forecasts feed the MRP engine.
2. **Sourcing:** Requisitions convert to RFQs and approved Purchase Orders.
3. **Receipt:** Inbound gate entry creates Gate Passes; warehouse receives and tests goods on GRN.
4. **Settlement:** Accepted items auto-post to stock; Finance executes 3-Way Matching against vendor bills.

OUTBOUND STREAM · ORDER-TO-CASH (O2C)

1. **Booking:** Sales Orders pass credit scrutiny & lock soft stock reservations.
2. **Production:** Deficits trigger work orders, consuming staged components.
3. **WMS Wave:** Pick waves direct mobile scanners; packing tables seal cartons.
4. **Logistics & AR:** TMS secures freight; e-Way Bills issue; dispatch triggers tax invoice.

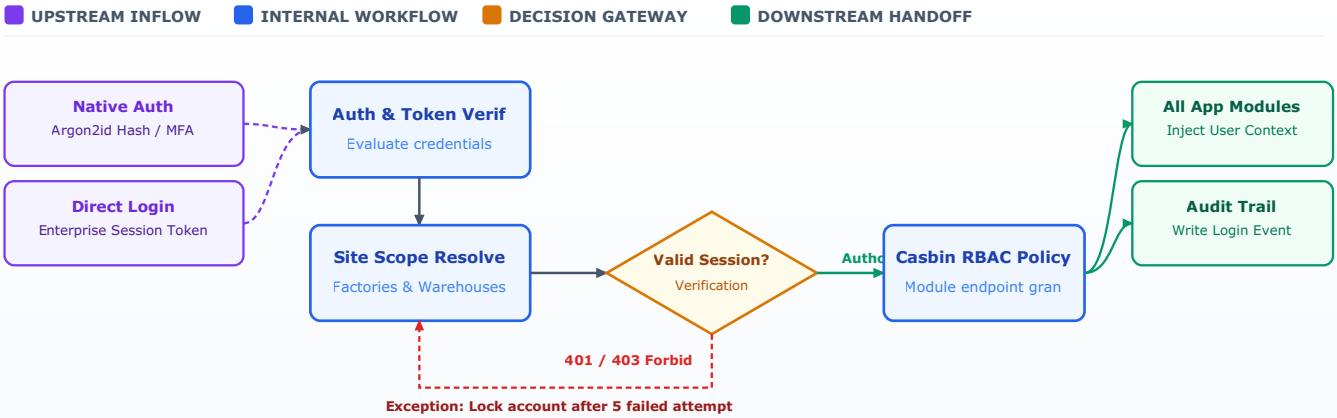
Module 01: Identity, Authentication & RBAC

/LOGIN, /PROFILE, /USERS

Entities: `users`, `roles`, `user_roles`, `user_role_warehouses`, `user_role_factories`, `sessions`

Primary Actors: System Admin, All Operational Users

Governs enterprise authentication, multi-factor token issuance, native Argon2id credentials, session lifecycle, and strict site-level data scoping across multiple factories and warehouses.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Authentication Handshake:** User provides email/password and optional MFA code. The backend verifies the password hash using Argon2id and validates credentials against active account policies.
- **MFA & Session Issuance:** If TOTP is active, pyotp validates the 6-digit passcode. A cryptographic session is minted in the `sessions` table with an expiration timestamp and stored in an HTTP-only secure cookie.
- **Site & Warehouse Scoping:** The user's assigned roles are loaded along with explicit warehouse (`user_role_warehouses`) and factory (`user_role_factories`) access bindings. If an operator is restricted to Warehouse WH-01, queries across inventory, GRN, and orders automatically inject tenancy filters.
- **Casbin Enforcement & Routing:** Casbin RBAC policy evaluates `(sub, obj, act)` on each API endpoint. If permitted, the request executes; otherwise, a 403 Forbidden response is returned and logged in `audit_log`.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Native Identity Engine	Argon2id credential store, multi-factor TOTP verification, and session manager.
Organization Admin	Assigns user records to specific business units, factories, and operational roles.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

All Functional Domains	Every API route depends on `get_current_user` dependency, propagating `user_id`, `role`, and `allowed_warehouse_ids`.
Audit Logging	Every state transition records `performed_by_user_id` for SOX compliance.

Lifecycle State Engine:

`USER: PENDING_ACTIVATION -> ACTIVE -> SUSPENDED -> TERMINATED | SESSION: ACTIVE -> REVOKED / EXPIRED`

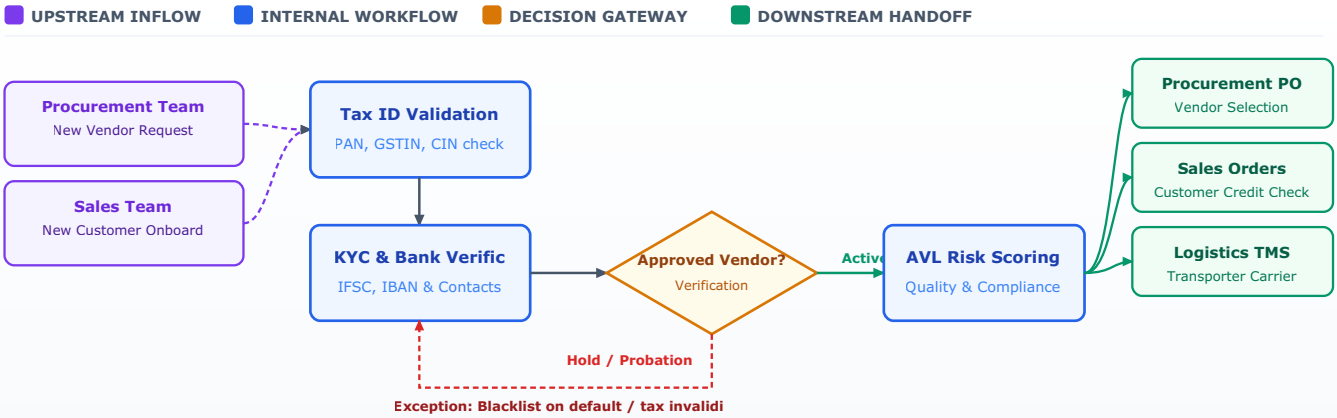
Module 02: Parties & AVL Governance

Entities: `parties`, `party_types`, `party_addresses`, `party_bank_accounts`, `party_versions`

/PARTIES, /PARTY/:PARTYID, /AVL-GOVERNANCE

Primary Actors: Procurement Manager, Sales Operations, Compliance Officer

Master registry for all external commercial entities: Vendors, Customers, Transporters, and Job Workers. Enforces Approved Vendor List (AVL) compliance, KYC/Tax ID validation, and credit limit guardrails.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Party Intake & Categorization:** Commercial entity created with a formal legal name, trading name, and designated party type (VENDOR, CUSTOMER, TRANSPORTER, JOB_WORKER).
- **Tax & Regulatory Validation:** System executes checksum and format validations on PAN (10 alphanumeric) and state GSTIN (15 characters). Cryptographic hashing ensures duplicate tax entities cannot be fraudulently registered.
- **Banking & Address Verification:** Multi-site billing and shipping addresses are recorded with postal codes and state codes. Bank accounts are verified for automated NEFT/RTGS AP settlement.
- **AVL Governance & Credit Scrutiny:** Procurement/Quality audits assign AVL status (`APPROVED`, `PROBATION`, `DISQUALIFIED`). For customers, a strict credit limit and payment terms (e.g. Net 30) are established. Only `APPROVED` vendors can be selected on Purchase Orders.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Foundation Domain	References `party_types` and currency codes.
Quality & Audit Domain	Supplier scorecard audits feed directly into the AVL rating calculation.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Procurement (PO)	Mandatory foreign key `purchase_orders.vendor_id` -> `parties.id`. Inactive or blocked parties reject PO creation.
Sales (Orders)	Foreign key `orders.customer_id` -> `parties.id`. Real-time query validates outstanding receivables against credit limit.
Logistics (TMS)	Transporters (`party_type = TRANSPORTER`) are invited to freight RFQs and bound to carrier dispatch manifests.
Finance (AP/AR)	Supplies bank coordinates for AP payouts and billing addresses for tax-compliant GST invoices.

Lifecycle State Engine:

```
PARTY: DRAFT -> UNDER_REVIEW -> APPROVED -> PROBATION -> BLOCKED -> ARCHIVED
```

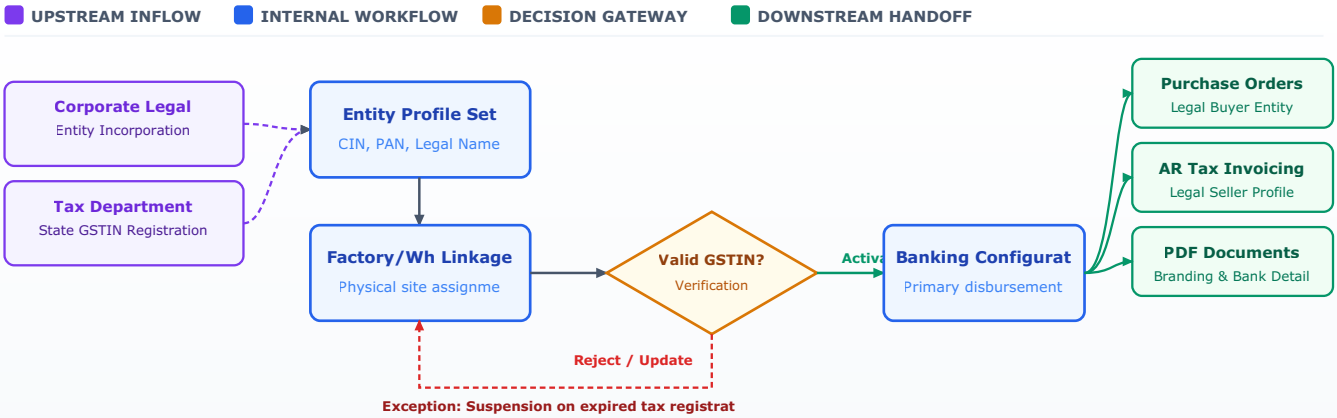
Module 03: Business Entities & Seller Profiles

/PO-INVOICE

Entities: `business_entities`, `factory_business_entities`, `business_entity_bank_accounts`

Primary Actors: Finance Director, Legal Counsel, Corporate Controller

Establishes corporate legal entities, state-level GSTIN registrations, registered headquarters, corporate seals, and primary banking coordinates for all buying and selling transactions.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Entity Registration:** Corporate identity recorded with Corporate Identification Number (CIN), Permanent Account Number (PAN), and state-specific Goods and Services Tax Identification Numbers (GSTIN).
- **Facility Association:** Operating factories and warehouses are mapped to corresponding legal business entities. This enforces correct intrastate vs interstate tax rules (CGST/SGST vs IGST) based on facility location.
- **Bank Disbursement Allocation:** Virtual and physical commercial bank accounts are linked to the business entity, designating primary clearing accounts for vendor electronic fund transfers.
- **Document Engine Branding:** Official digital signatures, corporate logos, registered office addresses, and default PO/ Invoice legal disclaimers are assigned for automated PDF rendering.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Foundation Domain	Configures parent corporate identity and currency standards.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Procurement (PO)	Every Purchase Order carries `business_entity_id`, defining which internal corporate entity is contractually liable for payment.
Finance (Invoicing)	AR invoices use the business entity's GSTIN and state code to compute state tax liability and print corporate bank details.
Logistics (e-Way Bill)	Supplies the consignor/consignee tax registration numbers for statutory transport compliance.

Lifecycle State Engine:

```
BUSINESS_ENTITY: DRAFT -> ACTIVE -> REVISED -> DEREGISTERED
```

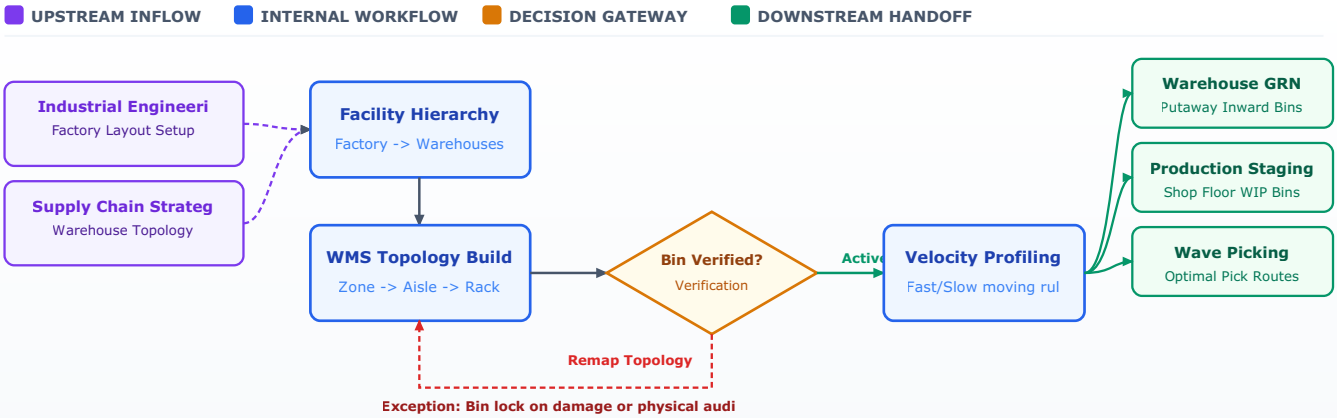
Module 04: Facilities, Factories & WMS Bins

/FACTORY-WAREHOUSE, /WMS-BINS

Entities: `factories, warehouses, locations (bins, racks, aisles, zones)`

Primary Actors: Plant Head, Warehouse Operations Manager, Logistics Engineer

Models the physical industrial topology: manufacturing factories, centralized warehouses, regional distribution centers, and granular WMS bin locations with capacity and velocity rules.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Factory Definition:** Manufacturing sites are established with geographical addresses, operating cost centers, labor rosters, and linked legal entities.
- **Warehouse Categorization:** Dedicated storage depots are created for specific functions: Raw Materials (RM), Work-in-Progress (WIP), Finished Goods (FG), and Quarantine / Scrap.
- **Multi-Tier WMS Hierarchy:** Each warehouse is subdivided into Zones (e.g. Ambient, Cold, High-Value), Aisles, Racks, Shelves, and individual Bins with volumetric and weight limits.
- **Velocity & Putaway Logic:** Fast-moving items are mapped to lower-tier, dock-adjacent bins to minimize forklift travel time during wave picking.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Identity (RBAC)	User site scoping filters bind warehouse workers to specific physical facilities.
Business Entities	Every warehouse and factory must belong to a valid corporate legal entity.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

GRN Receiving	Determines allowable receiving docks and default inbound inspection bins.
Inventory Balances	`stock_balances` and `stock_ledger` records maintain quantities broken down by `warehouse_id` and `location_id` (bin).
Production	Defines the staging locations for raw material shop floor issues and WIP queues.
Picking & Shipping	Wave picking generates shortest-path travel routes through aisles and racks.

Lifecycle State Engine:
`LOCATION: ACTIVE -> FULL -> LOCKED_FOR_AUDIT -> DAMAGED -> INACTIVE`

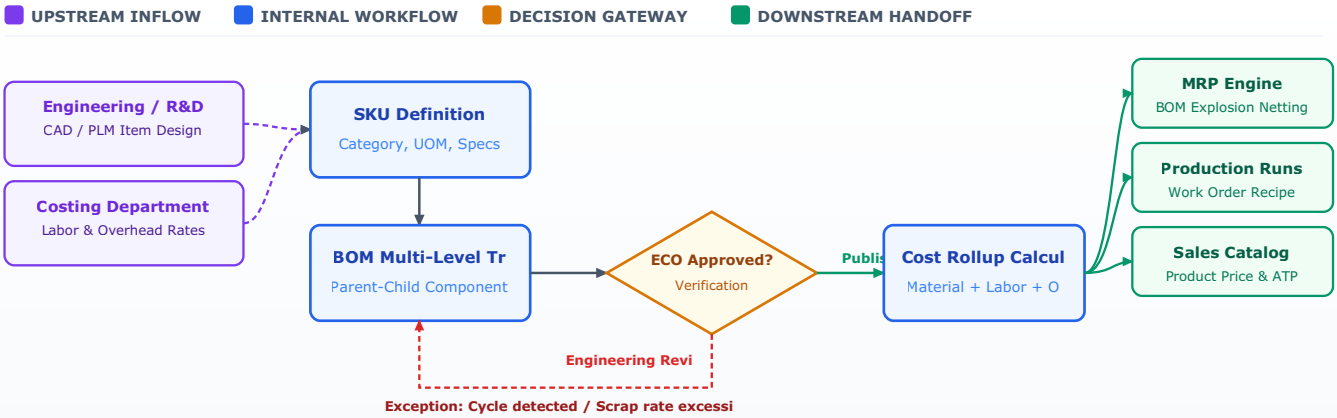
Module 05: Product Catalog & Engineering BOM

/PRODUCTS

Entities: `products`, `product_categories`, `boms`, `bom_versions`, `bom_components`, `bom_approvals`

Primary Actors: R&D Engineer, Product Manager, Cost Accountant

Defines sellable and component SKUs, engineering Bill of Materials (BOM) multi-level hierarchies, scrap allowances, version effectivity dates, and automated standard cost rollups.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **SKU Master Creation:** Item categorized as RAW_MATERIAL, SUB_ASSEMBLY, FINISHED_GOOD, or CONSUMABLE. System auto-generates unique SKU codes and barcodes with base and secondary UOM conversion ratios.
- **BOM Composition:** Engineers construct parent-child component trees specifying component part IDs, gross quantity per build unit, and planned scrap loss percentages.
- **Version Control & Effectivity:** BOMs are version-controlled (`v1.0`, `v2.0`). Engineering Change Orders (ECO) define effectivity start and end dates. Obsolete components are automatically blocked from future production.
- **Standard Cost Rollup Engine:** The system traverses the BOM hierarchy, aggregating active component purchase costs plus machine operation costs and direct labor rates to establish the standard manufacturing cost.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Product Categories	Inherits default HSN/SAC codes, tax rates, and inventory valuation methods (FIFO/Weighted Average).

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

MRP Engine	Explodes active BOMs to calculate gross-to-net raw material purchasing requirements.
Production Execution	Supplies the definitive component recipe for Work Order material staging and component consumption.
Procurement	Provides component specifications and authorized manufacturer part numbers.
Sales Catalog	Exposes finished goods SKUs for customer quoting and order entry.

Lifecycle State Engine:

```
BOM: DRAFT -> SUBMITTED -> PENDING_ENGINEERING_REVIEW -> APPROVED -> ACTIVE -> OBSOLETE
```

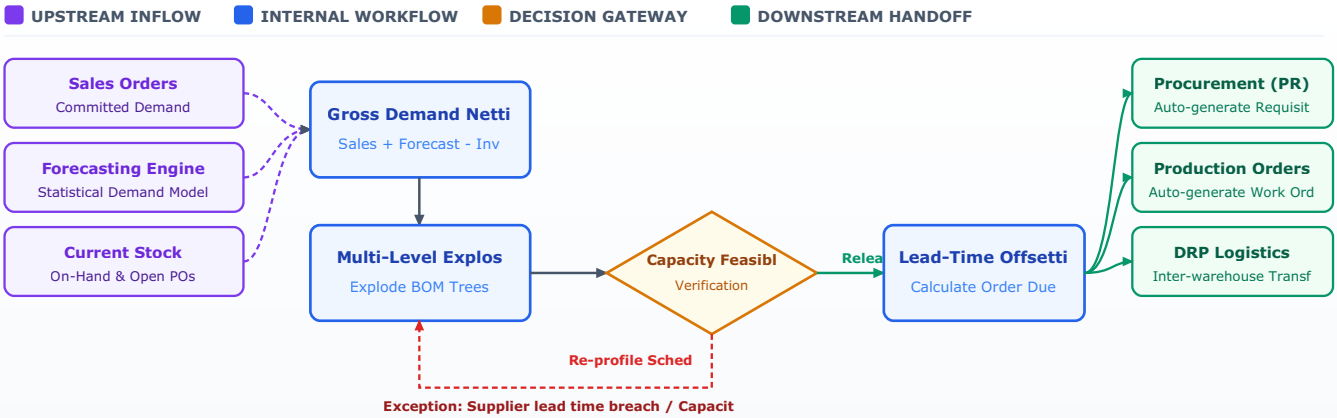
Module 06: Planning: Demand, MRP & DRP

/FORECASTING, /MRP, /DRP, /CAPACITY-PLANNING

Entities: `mrp_runs`, `mrp_planned_orders`, `drp_transfers`, `sales_forecasts`

Primary Actors: Master Scheduler, Supply Chain Planner, Inventory Controller

Synthesizes sales forecasts, committed orders, existing inventory, and open purchase orders to run multi-echelon MRP explosions and DRP replenishment across all factory and distribution hubs.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Demand Ingestion & MPS:** Aggregates committed Sales Orders and statistical sales forecasts into a time-phased Master Production Schedule (MPS).
- **Gross-to-Net Inventory Calculation:** Deducts currently available physical inventory, allocated stock, and scheduled in-transit PO receipts to derive net deficit for each finished product.
- **Multi-Level BOM Explosion:** Traverses product BOMs to explode component requirements, multiplying gross quantities by assembly batch size and applying scrap factors.
- **Lead Time Offsetting & Lot Sizing:** Applies vendor lead times and production cycle times backwards to determine release dates, applying Minimum Order Quantities (MOQ) and packaging multiples.
- **Planned Order Generation:** Emits automated Purchase Requisitions for raw materials and planned Work Orders for in-house manufacturing.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Sales Orders	Supplies hard unfulfilled sales lines.
Inventory Balances	Queries current unreserved stock balances and safety stock thresholds.
Product BOM	Provides component ratios and lead time attributes.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Requisitions Domain	Auto-creates records in `purchase_requisitions` for external supplier procurement.
Production Domain	Creates proposed manufacturing runs in `production_runs` (Planned status).
DRP Transfers	Creates inter-warehouse transfer requests to rebalance regional inventory.

Lifecycle State Engine:

```
MRP_RUN: SCHEDULED -> RUNNING -> COMPLETED -> ORDERS_GENERATED -> ARCHIVED
```

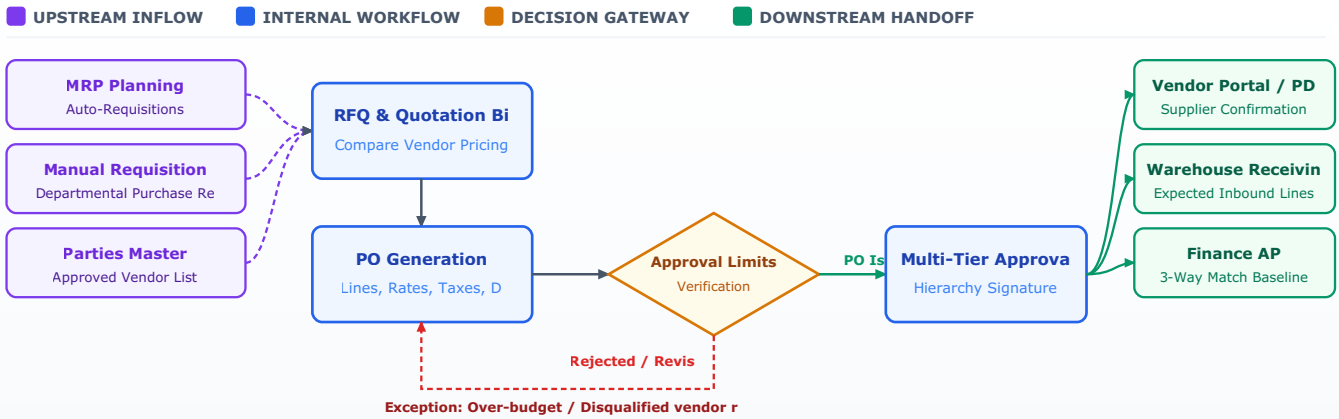
Module 07: Procurement, RFQ & Purchase Orders

/PROCUREMENT, /BLANKET-PO, /VENDOR-PORTAL

Entities: `purchase_requisitions`, `rfqs`, `purchase_orders`, `po_lines`, `po_approvals`, `blanket_pos`

Primary Actors: Buyer, Procurement Head, Finance Approver, External Vendor

End-to-end purchasing workflow: converts requisitions to RFQs, manages supplier bidding, issues formal Purchase Orders (Standard or Blanket), enforces approval hierarchies, and manages revisions.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Requisition & RFQ Bidding:** Approved Purchase Requisitions trigger RFQs distributed to AVL vendors. Vendor bids are ingested and compared for unit rate, payment terms, and delivery commitment.
- **Purchase Order Drafting:** Buyer selects awarded vendor and drafts PO specifying line items, HSN codes, tax structures, delivery schedules, freight terms, and recipient warehouse.
- **Multi-Tier Approval Hierarchy:** PO routes through dynamic approval gates based on total spend threshold (e.g. Level 1: Plant Buyer up to $5,000; Level 2: Supply Chain VP up to $50,000; Level 3: CFO beyond $50,000).
- **PO Dispatch & Vendor Acknowledgment:** Approved PO triggers background PDF compilation and email transmission to vendor, while publishing to the Vendor Portal for acknowledgment and ASN filing.
- **Amendment & Short-Close:** Changes generate versioned PO amendments (`PO-001-R1`). Buyers can short-close remaining unfulfilled balances when partial delivery is deemed final.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Requisitions / MRP	Pulls line item demand, quantity, and target delivery dates.
Parties Master	Validates vendor AVL status, currency, and tax IDs.
Business Entities	Defines the legal buyer identity on the PO header.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Inbound Logistics	Transporters reference PO numbers on Gate Passes.
Warehouse GRN	Inbound receiving matches vendor delivery against active PO line item tolerances.
Finance AP	Locked PO prices and line quantities form the immutable baseline for 3-Way Matching.
Supplier Scorecard	PO promised dates benchmark supplier On-Time In-Full (OTIF) ratings.

Lifecycle State Engine:

```
PO: DRAFT -> PENDING_APPROVAL -> APPROVED -> ISSUED -> PARTIALLY_RECEIVED -> RECEIVED -> BILLED -> CLOSED / SHORT_CLOSED
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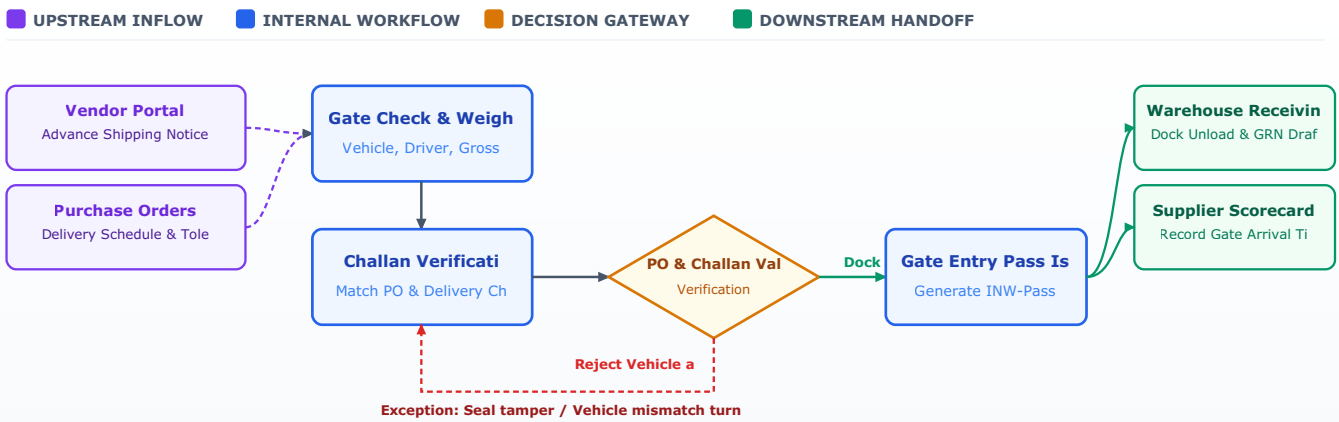
Module 08: Inbound Logistics & Gate Entry

/LOGISTICS, /FREIGHT-TMS

Entities: `transport_requests`, `gate_entries`, `dispatches`

Primary Actors: Security Gate Officer, Inbound Logistics Coordinator, Weighbridge Operator

Tracks inbound transport vehicles from vendor dispatch, processes Advance Shipping Notices (ASN), verifies driver credentials and Delivery Challans at the factory gate, and prints Gate Entry passes.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **ASN & Inbound Alert:** Vendor logs planned shipment in portal with transporter details, truck registration, LR/e-Way bill number, and estimated arrival date.
- **Physical Gate Arrival & Inspection:** Truck arrives at factory perimeter. Security inspects container seals, driver documentation, and matches physical Delivery Challan against open POs in the system.
- **Weighbridge Gross Weighing:** Truck drives onto automated weighbridge; gross vehicle weight is logged directly into the system to prevent bulk material theft or shorting.
- **Gate Entry Number Creation:** System generates a unique Gate Inward Pass (`INW-YYYY-XXXXX`), assigns a specific warehouse unloading dock, and releases the truck into the yard.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Procurement (PO)	Validates that active, approved PO lines exist for the incoming goods.
Transporter Master	Validates carrier identity and vehicle compliance records.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Warehouse GRN	Gate Entry Pass is mandatory foreign key to initialize a Goods Receipt Note.
Logistics Control	Tracks turnaround demurrage time from gate entry to final dock exit.

Lifecycle State Engine:

```
GATE_ENTRY: VEHICLE_ARRIVED -> INSPECTION_PASS ->
DOCK_ASSIGNED -> UNLOADING -> GATE_EXIT
```

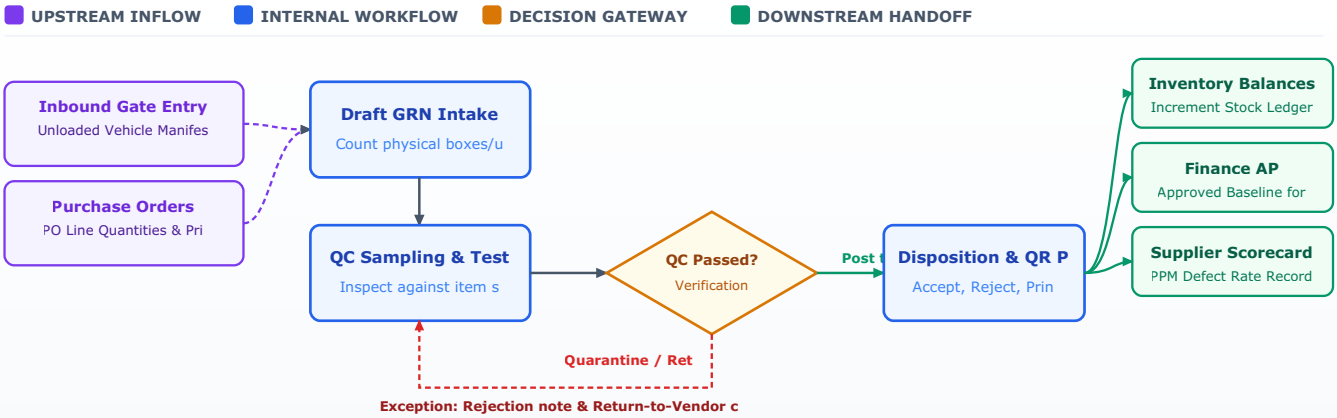
Module 09: GRN, Inbound QC & QR Labeling

/WAREHOUSE, /WAREHOUSE-QC/:GRNID

Entities: `grns, grn_lines, inventory_items, rejection_codes, sticker_sizes`

Primary Actors: Receiving Clerk, QC Inspector, Inventory Stockman

Manages physical material unloading, PO line quantity matching with over-receipt tolerances, rigorous QC sampling inspection, QR/Barcode sticker generation, and putaway execution.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Unloading & PO Matching:** Receiving team tallies physical packages against PO lines. System enforces over-receipt tolerance limits (e.g. max +5% allowed); excess quantities are blocked.
- **Draft GRN Creation:** Draft GRN records received quantity, vendor batch numbers, and manufacturing/expiry dates. Goods move to QC Inspection staging location.
- **Quality Control (QC) Inspection:** QC Inspector executes sampling protocol. Quality parameters are tested (dimensions, purity, electrical specs). Quantities are partitioned into `accepted_qty`, `rejected_qty`, and `quarantine_qty`.
- **QR/Barcode Label Generation:** Accepted units trigger automated generation of serialized QR sticker sheets (thermal roll 4x2, 2x1, or A4 24-up) containing SKU, GRN ID, batch ID, and tracking hash.
- **Putaway & Stock Ledger Posting:** Stockman scans QR labels and deposits pallets into assigned WMS warehouse bins. System atomically posts transactions to `stock_ledger` and updates PO received totals.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Purchase Orders	Links `grn.po_id` -> `purchase_orders.id` and updates line received quantities.
Gate Entry	Links inward vehicle pass for complete supply chain auditability.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Inventory Ledger	Creates active records in `inventory_items` and increments `stock_balances`.
Finance AP 3-Way Match	The `accepted_qty` on GRN lines defines the maximum payable quantity for vendor invoice clearance.
Supplier Scorecards	Rejection counts instantly degrade the vendor's quality scorecard rating.

Lifecycle State Engine:
`GRN: DRAFT -> PENDING_QC -> QC_IN_PROGRESS -> APPROVED -> COMPLETED / REJECTED`

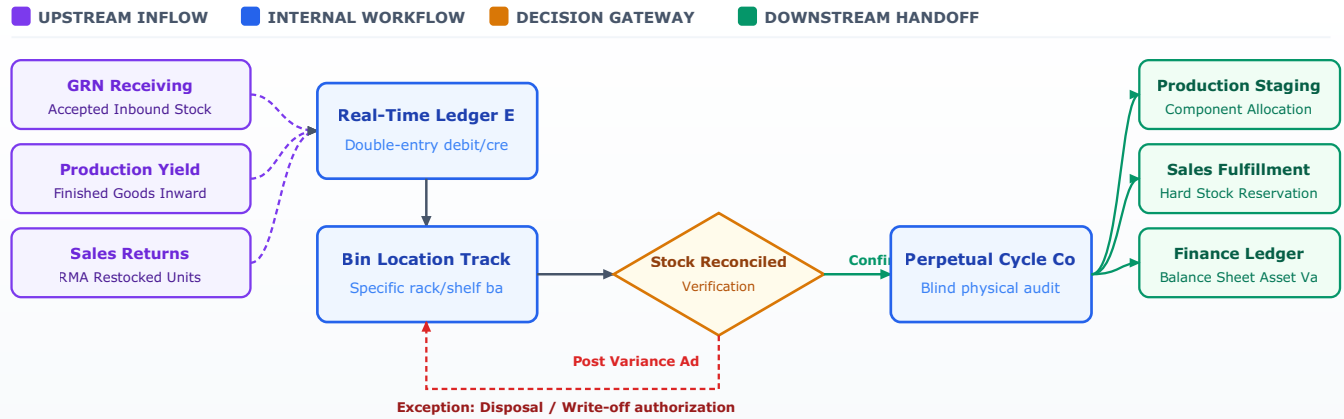
Module 10: Inventory Control & Stock Ledger

/INVENTORY, /INVENTORY-ITEM/:ITEMID, /CYCLE-COUNTS

Entities: `stock_balances, stock_ledger, inventory_items, transfers, cycle_counts`

Primary Actors: Warehouse Manager, Inventory Controller, Internal Auditor

Heart of the operational system: double-entry stock ledger, real-time bin balances, lot/serial traceability, inter-warehouse transfers, blind cycle counting, and scrap adjustments.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Double-Entry Stock Ledger:** Every material movement records an immutable row in `stock_ledger` with `quantity`, `uom`, `warehouse_id`, `location_id`, `valuation_rate`, and reference document.
- **Real-Time Stock Balances:** Queries return instant visibility into Total On-Hand, Allocated (Reserved for Production/Orders), In-Transit, and Net Available stock per SKU.
- **Bin Transfers & Rebalancing:** Warehouse staff scan source bin and destination bin to execute internal relocations without affecting overall facility balances.
- **Perpetual & Blind Cycle Counts:** Auditors initiate blind count batches where staff scan items without visibility to system expected quantities. Variances trigger supervisor investigation.
- **Inventory Adjustments & Disposals:** Authorized variances post debit/credit adjustments to the stock ledger with documented reason codes (damages, shrinkage, obsolescence).

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Warehouse GRN	Inward stock receipts.
Production Completion	Finished good and sub-assembly receipts.
Customer Returns (RMA)	Re-induction of restocked return goods.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Production Runs	Reserves components for active work orders; issues components to shop floor.
Sales Wave Picking	Locks inventory allocations and fulfills outbound sales orders.
Finance General Ledger	Feeds inventory asset valuations into corporate Balance Sheet accounts.

Lifecycle State Engine:

```
ITEM: IN_STOCK -> ALLOCATED -> PICKED -> SHIPPED | DISPOSED | UNDER_COUNT
```

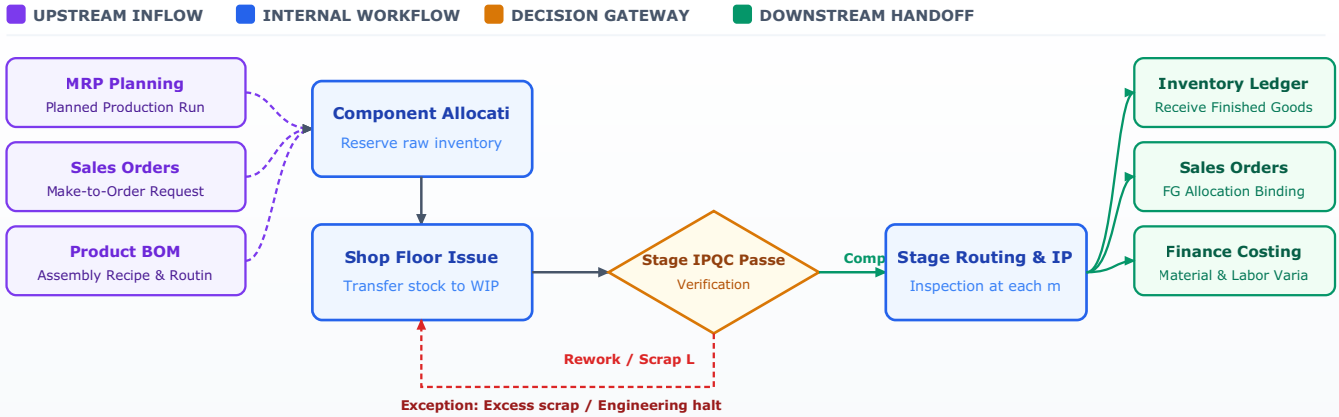
Module 11: Production Execution & IPQC

/PRODUCTION, /PRODUCTION/NEW, /PRODUCTION/:RUNID, /QUALITY-IPQC, / SUBCONTRACTING

Entities: `production_runs`, `production_consumptions`, `production_run_stages`, `finished_good_allocations`, `subcontracting_orders`

Primary Actors: Production Supervisor, Shop Floor Operator, IPQC Inspector

Transforms raw materials into finished assemblies: Work Order release, component staging & consumption, multi-stage routing, in-process quality gates, scrap tracking, and finished goods allocation.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Work Order Release & Allocation:** Production Order created for target finished SKU and batch quantity. System validates component availability against BOM and places hard reservation locks on raw materials.
- **Shop Floor Staging & Issue:** Materials transferred from main storage to shop floor WIP staging bins. System generates Material Transfer Slips with batch serialization.
- **Multi-Stage Execution & Routing:** Order progresses through sequential work center stages (e.g. SMT Assembly -> Soldering -> Enclosure Assembly -> Burn-in Testing).
- **In-Process Quality Control (IPQC):** At defined gate stages, IPQC inspectors perform testing. The order cannot advance to the next work center until the IPQC inspection report is digitally signed.
- **Subcontracting (Job Work):** If a stage requires external processing (e.g. powder coating), a Subcontracting Delivery Challan is issued, tracking material at vendor premises until inward return.
- **Yield Declaration & FG Allocation:** Final completion records accepted finished goods yield and scrap loss. Finished goods are immediately allocated to awaiting customer Sales Orders or posted to stock.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

BOM Domain	Inherits multi-level component requirements, standard scrap %, and routing stages.
Inventory Domain	Consumes raw material stock batches.
Sales Orders	Make-to-Order production runs are directly tagged with `order_id`.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Inventory Balances	Deducts raw materials from stock; adds newly minted finished good SKUs to FG warehouse.
Finished Good Allocations	Directly binds completed serials to waiting sales order lines.
Finance Costing	Computes actual manufacturing cost (actual component usage + labor/ overhead absorption) vs standard cost.

Lifecycle State Engine:

```
PRODUCTION_RUN: DRAFT -> PLANNED -> RELEASED ->
IN_PROGRESS -> IPQC_HOLD -> COMPLETED -> CLOSED
```

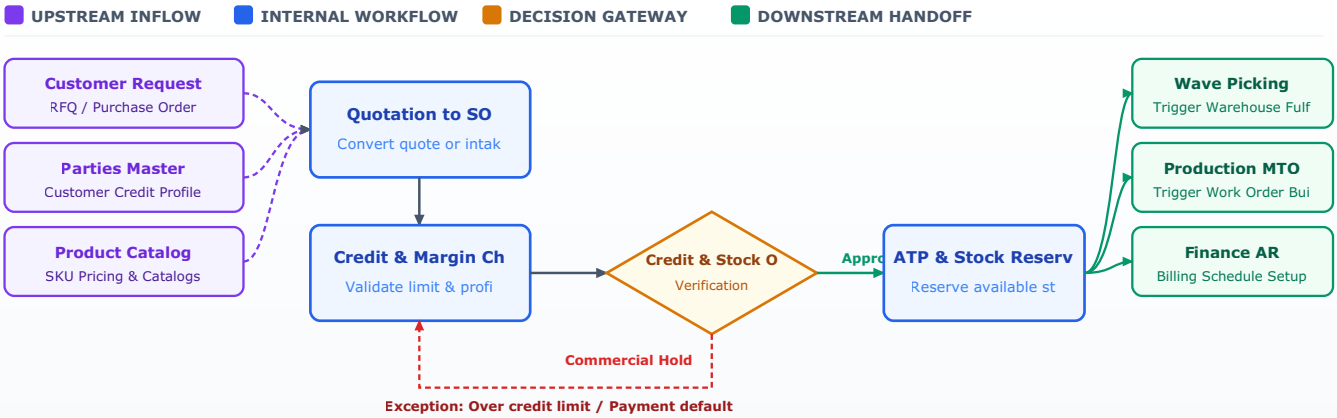
Module 12: Sales Management & Order Booking

/ORDERS, /CUSTOMER-PORTAL

Entities: `orders`, `order_lines`, `quotations`, `reservations`

Primary Actors: Sales Representative, Commercial Manager, Customer Service

Governs commercial revenue intake: customer quotations, sales order entry, automated credit limit checks, Available-to-Promise (ATP) stock netting, soft reservations, and fulfillment release.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Inquiry & Quotation:** Sales rep prepares quotation specifying line items, pricing tiers, discounts, delivery timelines, and payment terms.
- **Sales Order Intake:** Upon customer confirmation, quote converts to a binding Sales Order (`SO-YYYY-XXXXX`). Line items specify delivery addresses and requested dispatch dates.
- **Automated Credit & Financial Checks:** System queries the customer's total unpaid AR balance + unbilled open orders. If total exposure breaches the approved credit limit, the SO is placed on `CREDIT_HOLD`.
- **Available-to-Promise (ATP) & Soft Reservation:** System nets requested quantities against unreserved finished inventory. If stock exists, a soft reservation is created; if stock is deficient, a Make-to-Order production demand is triggered.
- **Order Release to Fulfillment:** Once approved, commercial lock is lifted and the order is published to the warehouse queue for wave picking.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Parties Master	Validates customer master, credit ceiling, tax exemption certificates, and addresses.
Products Master	Pulls sellable SKUs, pricing matrices, and tax rates.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Warehouse Wave Picking	Approved order lines appear on picking consoles for fulfillment.
Production Planning	Unstocked lines trigger make-to-order manufacturing runs.
Finance AR	Forms the contractual basis for generating tax invoices upon dispatch.

Lifecycle State Engine:

```
ORDER: DRAFT -> PENDING_CREDIT_CHECK -> CREDIT_HOLD -> APPROVED -> ALLOCATED -> PICKING -> SHIPPED -> INVOICED -> CLOSED
```

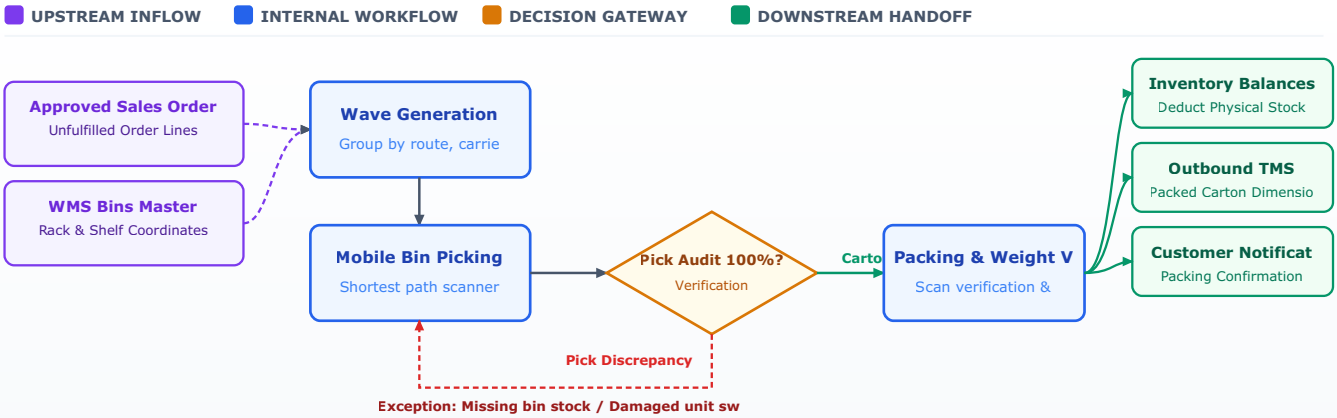
Module 13: Warehouse Wave Picking & Packing

/WAVE-PICKING, /WMS-MOBILE

Entities: `wave_picks, pick_tasks, packing_slips, cartons`

Primary Actors: Pick Wave Planner, Mobile Scanner Picker, Packing Station Lead

Optimizes order fulfillment: groups approved sales orders into intelligent pick waves, dispatches bin-directed tasks to mobile barcode scanners, verifies goods at packing tables, and generates carton labels.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Pick Wave Optimization:** Supervisor consolidates open sales orders into a wave based on shipping carrier, delivery zone, or high-priority SLA.
- **Bin-Directed Mobile Picking:** Warehouse pickers receive optimized traveling sequences on mobile handheld scanners. Staff must scan the bin barcode and SKU barcode to eliminate pick errors.
- **Packing Station Intake & Audit:** Picked items arrive in totes at packing stations. Pack lead scans each item to verify 100% quantity and quality conformity against order lines.
- **Cartonization & Barcode Labeling:** Items are packed into shipping cartons. System records carton dimensions, net weight, and tare weight, and prints carton shipping barcodes (`CARTON-001/003`).
- **Packing Slip & Staging:** System generates Packing Slip, seals cartons, and marks packages as `STAGED_AT_DOCK` for transporter pickup.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Sales Orders	Consumes approved order line item quantities.
WMS Facilities	Uses aisle and bin coordinates to compute shortest picking paths.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Inventory Stock Ledger	Converts soft reservations into definitive inventory issue deductions.
Outbound Logistics	Transfers carton count, volumetric dimensions, and gross shipping weight to the freight shipping engine.

Lifecycle State Engine:

```
WAVE_PICK: DRAFT -> RELEASED -> IN_PROGRESS -> PICKED -> PACKED -> STAGED
```

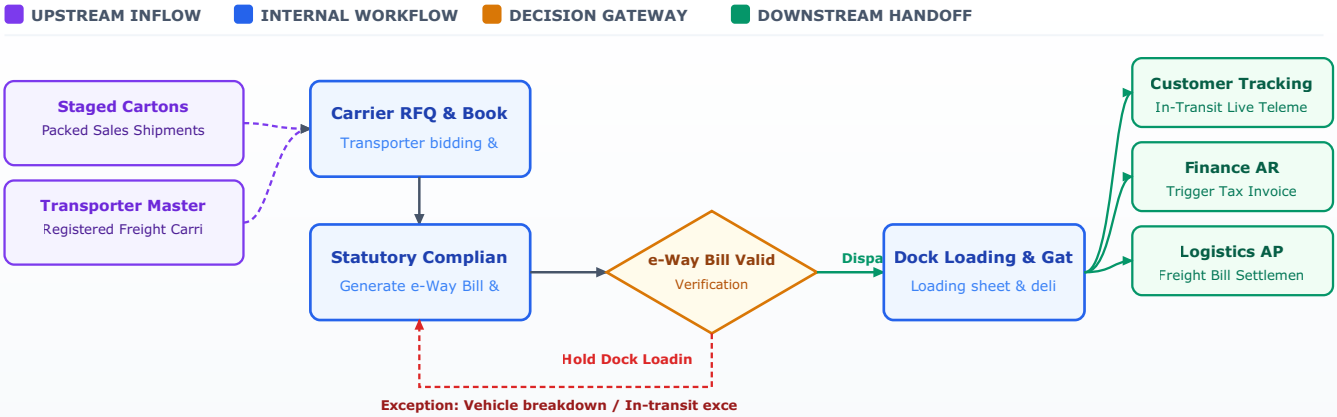
Module 14: Outbound Logistics & Freight TMS

/LOGISTICS, /FREIGHT-TMS, /STATUTORY-LOGISTICS, /SHIPPING-ENGINE

Entities: `dispatches, transport_requests, carrier_rfqs, carrier_bids, delivery_challans`

Primary Actors: Logistics Dispatcher, Transporter Carrier, Transport Accounts Clerk

Executes outward transport: runs transporter RFQ bidding, validates carrier vehicle placement, generates statutory e-Way Bills and e-Invoices, prints Delivery Challans, and tracks milestone PODs.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Freight RFQ & Transporter Bidding:** Dispatch request triggers spot RFQ or assigns primary contract transporter based on freight lane rate matrices.
- **Vehicle Placement & Dock Loading:** Assigned truck arrives at loading dock. Driver details and vehicle registration number are verified against booking records. Cartons loaded according to weight distribution.
- **Statutory Compliance Generation:** System connects to government GST/NIC portal API to auto-generate electronic e-Way Bill and electronic Tax Invoice (IRN QR code).
- **Delivery Challan & Gate Outward Pass:** Official Delivery Challan and Security Gate Outward Pass are issued. Truck exits perimeter; shipment status updates to `IN_TRANSIT`.
- **Milestone Tracking & POD Upload:** GPS milestones update shipment progress. Upon delivery, driver obtains signed physical or digital Proof of Delivery (POD) uploaded to the system.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Warehouse Packing	Receives total carton count, pallet volume, and shipping weight.
Business Entities	Pulls consignor legal entity details and state GST registration.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Sales Orders	Transitions sales order status to `SHIPPED` and `DELIVERED`.
Finance AR	Confirmed dispatch triggers final AR Tax Invoice generation.
Finance AP	Transporter POD serves as mandatory verification to clear freight bills for payment.

Lifecycle State Engine:

```
DISPATCH: READY_FOR_DISPATCH -> VEHICLE_ASSIGNED -> LOADED -> GATE_OUT -> IN_TRANSIT -> DELIVERED -> SETTLED
```

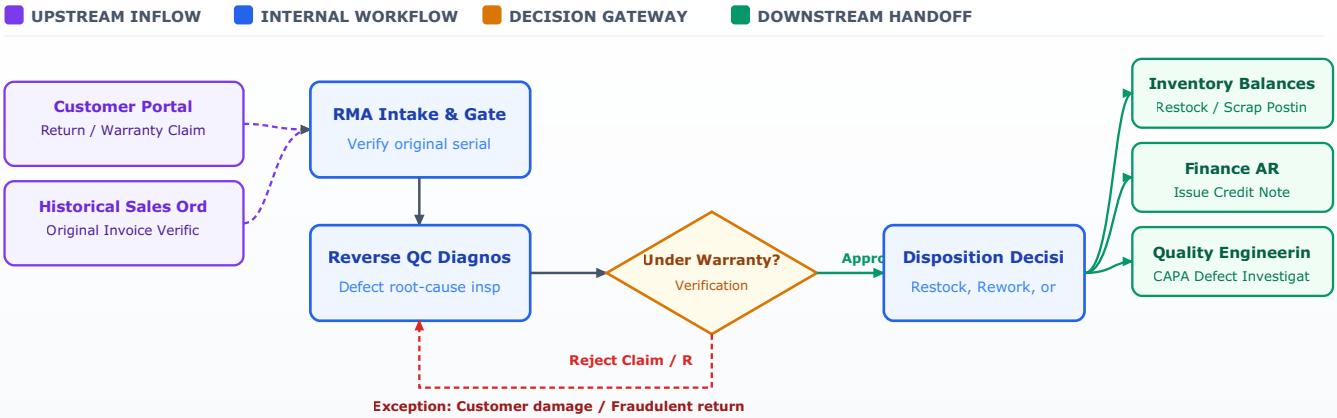
Module 15: Customer Returns & Reverse Logistics

/RETURNS

Entities: `returns`, `return_lines`, `return_inspections`, `credit_notes`

Primary Actors: Customer Support, Reverse QC Engineer, Inventory Restock Clerk

Manages customer Return Material Authorizations (RMA), reverse logistics intake, technical diagnostics, dispositioning (restock, rework, scrap), and financial credit note or replacement order issuance.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **RMA Initiation & Validation:** Customer submits return request. System cross-references original Sales Order and invoice to ensure warranty validity and return window eligibility.
- **Reverse Pickup & Gate Inward:** Reverse logistics pickup scheduled. Upon arrival at factory, security logs Return Gate Pass and delivers goods to Quarantine Returns staging.
- **Technical Diagnostics & Reverse QC:** Quality engineers test returned unit, verifying serial numbers and classifying root cause (transit damage, manufacturing defect, user error).
- **Dispositioning Execution:** Accepted units are dispositioned: (1) Good units sanitized and returned to active stock; (2) Fixable units routed to rework production; (3) Non-salvageable units scrapped.
- **Financial & Commercial Resolution:** Customer is resolved via: (A) Automated Credit Note issued in Finance AR against original invoice; or (B) Zero-dollar replacement Sales Order dispatched.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Sales Orders	Validates original purchase price, serial number validity, and customer identity.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Inventory Ledger	Increments warehouse stock if restocked, or logs scrap disposal.
Finance AR	Triggers formal GST-compliant Credit Note reducing customer receivables.
Quality Management	Feeds defect analytics into Corrective and Preventive Action (CAPA) dashboards.

Lifecycle State Engine:

```
RMA: REQUESTED -> RMA_APPROVED -> RECEIVED ->
QC_INSPECTION -> RESTOCKED / SCRAPPED -> SETTLED
```

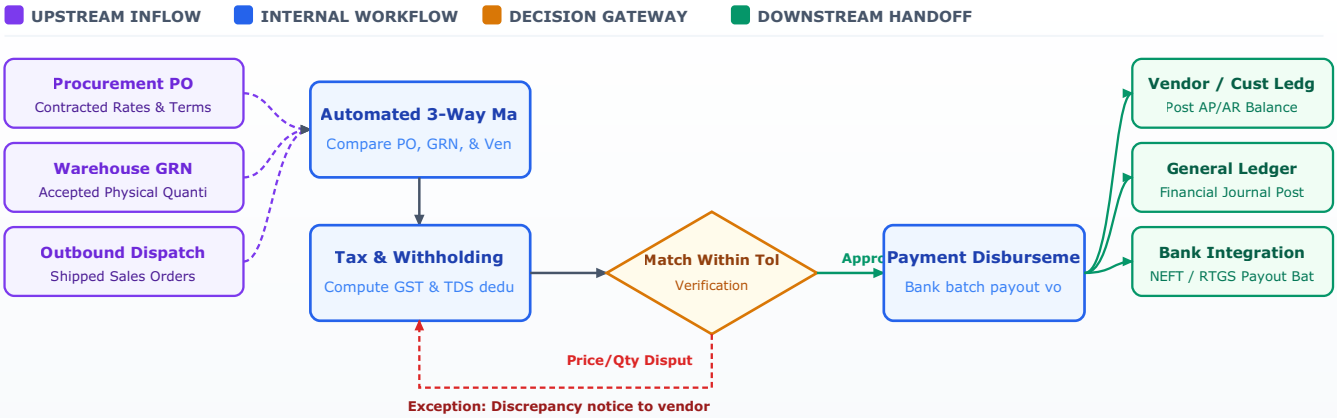
Module 16: Finance, Invoicing & 3-Way Match

/FINANCE

Entities: `invoices, invoice_lines, ap_invoices, ap_invoice_lines, payments, accounting_periods`

Primary Actors: Accounts Payable Officer, Accounts Receivable Lead, Chief Financial Officer

Financial nucleus of the ERP: automated 3-Way Matching (PO vs GRN vs Vendor Bill), AP payment vouchers, AR tax invoice issuance, tax withholding (TDS/GST), and accounting period governance.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **AP Vendor Bill Ingestion:** Vendor tax invoice submitted via portal or accounts payable desk. System parses line items, tax numbers, and invoice totals.
- **Automated 3-Way Matching Engine:** System executes rigorous 3-way check: (1) PO Unit Price vs Invoice Unit Price; (2) GRN Accepted Quantity vs Invoiced Quantity; (3) PO Payment Terms. Tolerances (e.g. max 1% price variance) are strictly enforced.
- **Invoice Clearance & Tax Booking:** Matched invoices auto-approve. System computes statutory withholdings (TDS under Income Tax Act, GST TDS) and books debit to Expense/ Inventory Asset and credit to Accounts Payable.
- **AR Tax Invoicing:** Outbound dispatches auto-generate AR Tax Invoices with sequential fiscal year numbering, posting debit to Customer Receivables and credit to Sales Revenue and GST Output Liability.
- **Payment Vouchers & Reconciliation:** Treasury runs payment batch generation, exporting bank clearing files. Incoming customer receipts are reconciled against open AR invoices.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Procurement	Pulls contracted rates, approved PO lines, and payment terms.
Warehouse GRN	Pulls verified accepted goods receipt lines.
Sales Dispatch	Pulls confirmed shipped order lines for AR billing.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Parties Ledger	Updates real-time outstanding vendor payables and customer credit utilization balances.
Accounting Periods	Restricts transaction posting to open accounting fiscal periods.

Lifecycle State Engine:

```
INVOICE: DRAFT -> PENDING_MATCH -> MATCH_FAILED_HOLD -> APPROVED -> SCHEDULED_FOR_PAYMENT -> PAID
```

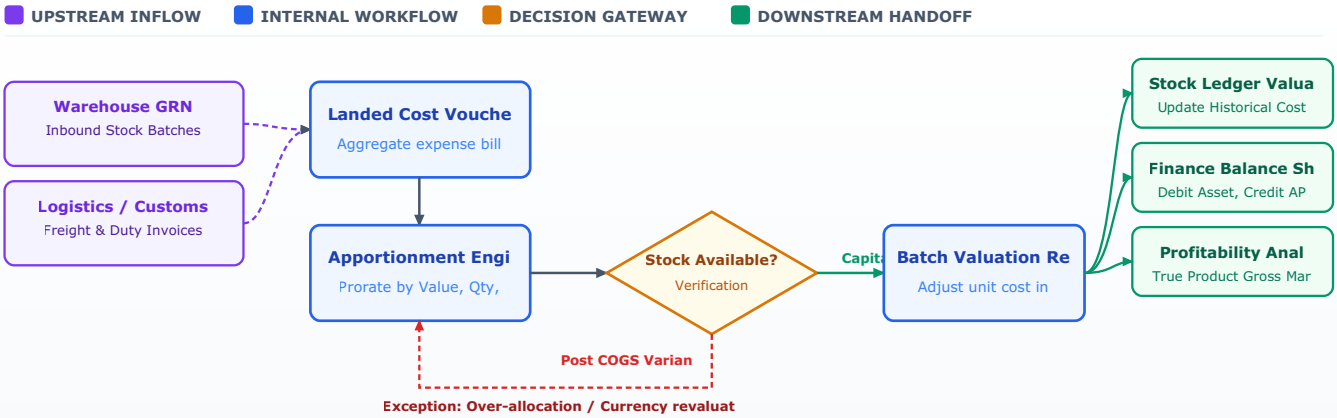
Module 17: Landed Cost & Inventory Valuation

/LANDED-COST

Entities: `landed_costs`, `landed_cost_items`, `landed_cost_allocations`, `stock_ledger`

Primary Actors: Cost Accountant, Import/Export Manager, Financial Analyst

Capitalizes indirect procurement expenses (customs tariffs, maritime freight, port handling, transit insurance) across physical inventory batches to reflect true landed cost in balance sheet valuation.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Expense Ingestion:** Third-party freight forwarder bills, customs clearance duties, and insurance invoices are booked and linked to an Inbound Shipment file.
- **Landed Cost Voucher Creation:** Cost accountant initiates Landed Cost Voucher (`LCV-YYYY-XXXXX`), attaching relevant inbound GRN batches.
- **Apportionment Calculation:** System distributes total expense across received items based on chosen allocation rule: By Value (ad valorem), By Quantity, By Net Weight, or By Cubic Volume.
- **Stock Ledger Revaluation Posting:** System calculates cost delta per unit. If inventory is still in warehouse, the unit valuation rate in `stock_ledger` is capitalized upward. If already sold, variance posts directly to Cost of Goods Sold (COGS).

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

Warehouse GRN	Supplies received item line quantities, initial PO purchase rates, and batch IDs.
Logistics / Freight Invoices	Supplies payable expense amounts from third-party logistics service providers.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

Inventory Balances	Increases balance sheet valuation of raw materials and purchased components.
Product Costing	Feeds actual landed cost baselines back into BOM engineering cost comparison.

Lifecycle State Engine:

```
LANDED_COST: DRAFT -> EXPENSES_LINKED -> ALLOCATED -> POSTED_TO_LEDGER
```

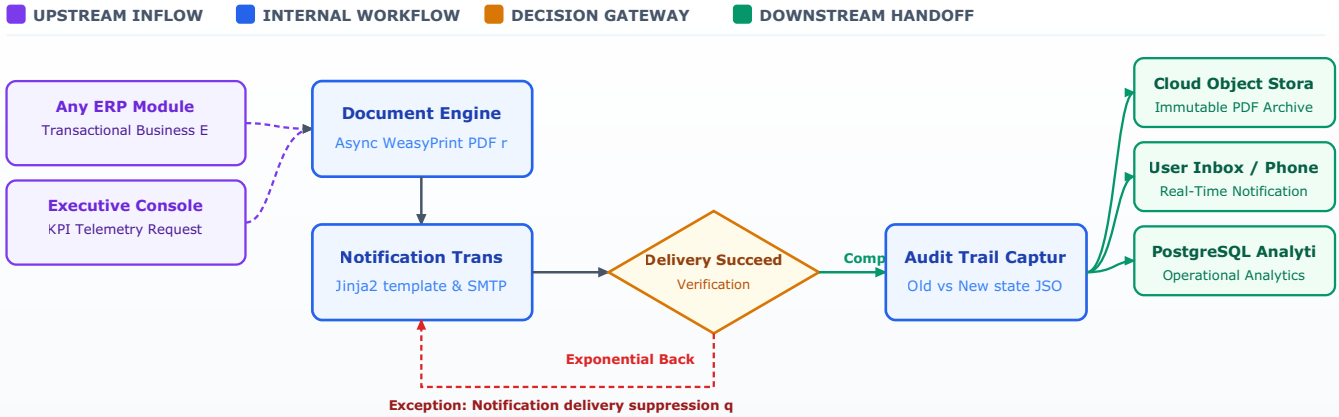
Module 18: Platform Engines & Control Tower

/DOCUMENTS, /NOTIFICATIONS, /AUDIT, /SC-CONTROL-TOWER

Entities: `documents, notifications, notification_templates, audit_logs, delivery_logs`

Primary Actors: System Administrator, Executive Operations, Compliance Auditor

Cross-cutting foundation: asynchronous headless PDF rendering, multi-channel transactional alerts (Email, SMS, Webhook), immutable regulatory audit logging, and the executive Supply Chain Control Tower.



WORKFLOW EXECUTION LOGIC & BUSINESS RULES

- **Asynchronous Document Generation:** Transactions (PO approval, GRN receipt, Tax Invoice, e-Way Bill) emit document render events. Background worker executes WeasyPrint to compile pixel-perfect PDFs, uploads to S3/Cloud Storage, and returns secure pre-signed URLs.
- **Multi-Channel Notification Dispatch:** Event triggers match configured templates, substituting runtime variables. Messages are dispatched via SMTP email, SMS gateway, or webhook with automatic retry backoff.
- **Immutable Audit Logging:** Every mutating REST API request captures `user_id`, `ip_address`, `endpoint`, timestamp, entity name, entity ID, and a JSON diff of pre-change and post-change state. Audit records are tamper-proof.
- **Supply Chain Control Tower:** Aggregates cross-module metrics in real-time: Supplier OTIF, Inventory Turn Ratio, WIP Shop Floor Aging, Open Order Fill Rate, and Cash Conversion Cycle, leveraging high-performance PostgreSQL queries for business intelligence.

CROSS-MODULE INTEGRATION CONTRACT

UPSTREAM DEPENDENCIES (INFLOWS)

All Functional Domains	Receives transactional triggers, documents to print, and audit state deltas.

DOWNSTREAM INTEGRATIONS (OUTFLOWS)

External Stakeholders	Delivers official signed documents and email alerts to suppliers and customers.
PostgreSQL Operational Store	Maintains operational transactional records and aggregated materialized metrics for enterprise reporting.

Lifecycle State Engine:

`DOCUMENT: QUEUED -> RENDERING -> STORED | NOTIFICATION: QUEUED -> SENT -> FAILED`

Cross-Module Master Integration Matrix • Part I: Inbound & Manufacturing

This master matrix maps explicit cause-and-effect data relationships across domain boundaries when key business events occur during foundation setup, planning, procurement, receiving, and production.

SOURCE MODULE & TRIGGER	BUSINESS EVENT	TARGET MODULE	IMPACTED ENTITY & STATE CHANGE	OPERATIONAL & FINANCIAL IMPACT
M01: Identity Login / Session	User authenticates via native credentials or MFA	**All Application Domains**	`sessions.is_active = True` Injects `allowed_warehouses`	Enforces row-level site scoping; operators cannot view or manipulate inventory outside authorized facilities.
M02: Parties Vendor Blacklist	Vendor fails audit or tax registration lapses	**M07: Procurement & M16: Finance**	`parties.status = 'BLOCKED'` `party.avl_rating = 'DISQUALIFIED'`	Blocks creation of new Purchase Orders; freezes outstanding AP payment voucher disbursement pending audit.
M03: Entities GSTIN Registration	New state tax branch entity approved	**M07: Procurement & M14: Logistics**	`business_entities.status = 'ACTIVE'` Linked to local factory	Governs intrastate (CGST+SGST) vs interstate (IGST) tax calculation; populates consignor in e-Way bills.
M04: Facilities WMS Bin Lock	Bin physical damage or audit freeze	**M09: GRN Putaway & M13: Wave Picking**	`locations.is_locked = True`	Prevents putaway tasks from routing to this bin; redirects pick wave algorithms around locked storage aisles.
M05: Products BOM Revision Lock	Engineering releases approved BOM v2.0	**M06: Planning (MRP) & M11: Production**	`bom_versions.status = 'ACTIVE'` Old version marked `OBSOLETE`	MRP explodes new component tree; subsequent Work Orders draw revised part quantities; obsoleted parts blocked.
M06: Planning MRP Run Complete	Time-phased net deficit calculation finishes	**M07: Procurement & M11: Production**	Inserts rows into `purchase_requisitions` Inserts rows into `production_runs`	Generates planned purchase requisitions with vendor lead-time offsetting; drafts manufacturing orders for assembly lines.
M07: Procurement PO Approval	Final signature limit sign-off approved	**M08: Logistics & M09: GRN & M16: Finance**	`purchase_orders.status = 'APPROVED'` `po_lines.is_locked = True`	Publishes order to Vendor Portal; opens receiving window at warehouse gate; locks price ceiling for 3-way match.
M08: Logistics Gate Entry Pass	Inbound truck arrives with vendor shipment	**M09: Warehouse Receiving**	`gate_entries.status = 'DOCK_ASSIGNED'` Generates `INW-YYYY-XXXXX`	Permits receiving staff to initialize Draft GRN linked to confirmed gate manifest; starts dock dwell time tracking.
M09: Warehouse GRN QC Approval & Putaway	Sampling inspection accepts incoming batch	**M10: Inventory & M16: Finance & M02: Supplier AVL**	`grns.status = 'COMPLETED'` Inserts into `stock_ledger` & `inventory_items`	Increments real-time stock balance; generates QR stickers; registers maximum payable quantity for AP 3-Way Match.

Cross-Module Master Integration Matrix • Part II: Outbound & Finance

This section maps downstream cause-and-effect relationships during production completion, order allocation, warehouse wave picking, freight transportation, customer returns, and financial settlement.

SOURCE MODULE & TRIGGER	BUSINESS EVENT	TARGET MODULE	IMPACTED ENTITY & STATE CHANGE	OPERATIONAL & FINANCIAL IMPACT
M10: Inventory Cycle Count Variance	Auditor posts blind count reconciliation	**M16: Finance (GL) & M18: Audit Trail**	`stock_ledger` adjustment posted Creates journal voucher	Updates physical inventory balances to match truth; posts inventory shrinkage or surplus variance to P&L expense.
M11: Production Run Completion	Assembly finishes; IPQC passes yield	**M10: Inventory & M12: Sales Orders & M16: Finance**	Inserts FG into `inventory_items` Updates `finished_good_allocations`	Increments finished goods stock; binds serials to waiting Make-to-Order Sales Orders; posts labor & overhead cost variances.
M12: Sales Orders Order Booking	Customer signs commercial quote	**M02: Parties & M10: Inventory & M13: Wave Picking**	`orders.status = 'APPROVED'` Inserts into `reservations`	Deducts available credit limit; places soft allocation on inventory; queues order lines on warehouse fulfillment consoles.
M13: Wave Picking Packing Complete	Cartons packed, weighed, and sealed	**M10: Inventory & M14: Outbound TMS**	`wave_picks.status = 'PACKED'` Decrements physical `stock_balances`	Permanently issues inventory out of warehouse; sends package dimensions and weight to transport shipping engine.
M14: Logistics Goods Gate Outward	Vehicle leaves with e-Way Bill & Challan	**M12: Sales Orders & M16: Finance (AR)**	`orders.status = 'SHIPPED'` Auto-generates `ar_invoices`	Transitions Sales Order to shipped; automatically generates tax invoice with GST liability; updates customer ledger.
M15: Returns RMA QC Disposition	Returned unit inspected & approved	**M10: Inventory & M16: Finance (AR)**	Restocks unit to `stock_balances` Inserts into `credit_notes`	Re-inducts sanitized item into good inventory bin; creates official Credit Note reducing customer outstanding debt.
M16: Finance AP 3-Way Match Pass	PO, GRN, and Vendor Invoice align	**M02: Parties Ledger & M18: Bank Integration**	`ap_invoices.status = 'APPROVED'` Queues payment in `payments`	Credits accounts payable ledger; queues disbursement in upcoming bank NEFT/ RTGS payment run.
M17: Landed Cost LCV Voucher Post	Customs and maritime freight allocated	**M10: Inventory Balances & M16: Finance GL**	Updates `stock_ledger.valuation_rate` Debits Inventory Asset Account	Capitalizes shipping duties into true inventory cost; adjusts balance sheet asset valuation; refines gross margin accuracy.
M18: Platform Document Generated	Async PDF render completes	**External Parties & Documents Archive**	Updates `documents.status = 'READY'` Dispatches email notification	Delivers digitally signed PDF copy to supplier/customer inbox; archives encrypted copy in immutable S3 storage.

End-to-End Enterprise Value Streams & Governance

In enterprise operations, business activities cross multiple domain modules concurrently. Below are detailed architectural walkthroughs for the two primary commercial engines of the system.

1. THE PROCURE-TO-PAY (P2P) COMPLETE OPERATIONAL JOURNEY

Phase 1: Demand & Sourcing (M06 → M07 → M02): The MRP engine nets sales demand against stock and emits a Requisition. Buyer initiates an RFQ to Approved Vendor List (AVL) suppliers. The lowest qualified bid converts into an official Purchase Order. Approval rules require Director sign-off based on spend limits. The PO is digitally issued to the vendor.
Phase 2: Inbound Delivery & Gate Verification (M08 → M09): The vendor ships goods with an ASN. Upon arrival, Security verifies vehicle seals, records gross weighbridge tonnage, and mints an Inward Gate Entry Pass. The truck unloads at the assigned receiving dock.
Phase 3: Receipt, QC & Inventory Induction (M09 → M10 → M18): Receiving verifies PO lines against over-receipt tolerance. QC pulls test samples; upon approval, thermal QR stickers are printed and affixed. Forklifts put away pallets into WMS bins. Stock ledger records the receipt, increasing available inventory.
Phase 4: Financial Matching & Disbursement (M16 → M17 → M02): Vendor submits electronic bill. System executes 3-Way Matching: PO unit price vs Invoice price, GRN accepted quantity vs Invoiced quantity. If matching succeeds within tolerance, AP invoice approves. Landed cost freight vouchers capitalize into inventory value. Bank payment voucher disburses funds.

2. THE ORDER-TO-CASH (O2C) COMPLETE OPERATIONAL JOURNEY

Phase 1: Intake & Credit Clearance (M02 → M12 → M03): Customer submits commercial order. System checks party credit ceiling: total unpaid AR + open orders cannot exceed the limit. Pricing engine applies volume tiers and GST rates based on seller business entity.
Phase 2: Inventory Allocation & Manufacturing Trigger (M12 → M10 → M11): System evaluates Available-to-Promise (ATP). Available inventory is locked with a hard reservation. Any inventory deficit automatically triggers a Work Order in Production, pulling BOM recipes and staging raw materials.
Phase 3: Warehouse Wave Picking & Packaging (M13 → M10 → M04): Fulfillment generates pick waves. Mobile RF scanners guide pickers along shortest bin paths. Packing tables scan-verify 100% item accuracy, pack into shipping cartons, capture gross scale weights, and print carton barcodes.
Phase 4: TMS Dispatch, Statutory e-Way Bill & Invoicing (M14 → M16 → M18): Transporter RFQ secures freight carrier. System calls government API to generate electronic e-Way Bill and Tax Invoice (IRN). Gate officer verifies outward pass. Upon delivery, driver uploads signed Proof of Delivery (POD), completing the commercial transaction.

SYSTEM RELIABILITY, ACID TRANSACTIONS & POSTGRESQL ANALYTICS

• **ACID Transaction Isolation:** Multi-table operations (e.g., GRN posting creating stock ledger entries, inventory items, and updating PO line balances) execute inside atomic database transactions (`session.begin()`). Any mid-flight failure rolls back all mutations completely.
• **Idempotency Enforcement:** All mutation endpoints support `Idempotency-Key` headers stored in Redis/PostgreSQL, preventing double-posting of receipts, invoices, or payments during network timeouts.
• **High-Performance PostgreSQL Analytics:** Operational transactional tables leverage optimized composite indexing, PostgreSQL window functions, and real-time aggregation views, empowering executive Control Tower BI dashboards without impacting OLTP database performance.